#HW 11

Graded

Student

Eric Jacobsen

Total Points

98 / 100 pts

Question 1

(no title) **30** / 30 pts

✔ **− 0 pts** Correct

Question 2

(no title) **8** / 10 pts

− 0 pts Correct

− 1 pt for part b, the process can be possible as entropy transfer can be less than 0.

− 1 pt c, will be impossible. To make the entropy change 0 when the entropy transfer is more than 0, you need to have a negative entropy production, which is not possible.

✔ **− 1 pt** d will be indeterminate because if entropy change is less than the entropy transfer then energy production can become negative which is not possible. Only when the entropy change is greater than or equal to the entropy transfer, the entropy production will be greater than or equal to 0 which makes the process possible.

✔ **− 1 pt** g will be indeterminate too.

− 1 pt e will be possible.

− 5 pts It has not been explained whether the process will be possible or impossible or indeterminate.

− 1 pt f will be impossible since entropy production should always be positive.

− 1 pt a will be possible because in absence of entropy transfer, the entropy production must be positive to make entropy change greater than 0.

Question 3

(no title) **15** / 15 pts

✔ **− 0 pts** Correct

Question 4

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

- **– 2 pts** Part c will be irreversible since, Qh will be 1000 and as a result efficiency will be 60 percent.
- **– 2 pts** Part a will be irreversible since efficiency will be 0.5, which is less than 2/3.
- **– 2 pts** b will be reversible since efficiency will be 2/3. The work will be 600 kJ.
- **– 5 pts** You have to mention which process is reversible, which is irreversible, and which is impossible.

Question 5

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

- **– 2 pts** For efficiency calculation, temperatures should always be converted to Kelvin or Rankine.
- **– 2 pts** In part b, efficiency will be around 0.6. Answers will be different
- **– 1 pt** the answer of c is not clear.
- **– 2 pts** In part d Th is 1000 and you need to find Tc.

Question 6

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

- **– 1 pt** The efficiency calculation is wrong.

No questions assigned to the following page.

100g of solid copper is reacted with chlorine gas to form $CuCl_2$ ...
Mass of product is 200g ...

HW #11

2) 6.17.3

Using $\Delta S = S_{in} - S_{out} + S_{prod}$

a) Positive entropy change with no entropy transfer needs $S_{prod} > 0$
Possible

b) If $S_{prod} > 0$ and $\Delta S < 0$, then entropy change must be negative
Possible

c) If entropy transfer > 0 and $\Delta S = 0$, then S_{prod} must be negative which can't happen
Impossible

d) If $\Delta S > 0$ and entropy transfer > 0, S_{prod} must be positive
Possible

e) If $\Delta S > 0$ and entropy transfer < 0, S_{prod} must be positive
Possible

f) $S_{prod} < 0$ can't happen
Impossible

g) If $\Delta S < 0$ and entropy transfer < 0, then $S_{prod} \geq 0$
Possible

3a) 6.17.9

- T_H is the temperature of the hot reservoir, and T_C is temp of cold
$T_C < T_H$, so Q goes $T_H \longrightarrow T_C$
Reversing it would move heat from cold to hot with no work input, which if that could be done, could pair with a heat engine and convert all heat from the hot reservoir into work. That violates Kelvin-Planck which says a device can't turn all heat from a single reservoir into work

3b) If free expansion into a vacuum were reversible, gas could be compressed back to its original volume without requiring work or heat transfer. A combo of this "free compression" with a normal heat engine would allow a conversion of all heat from a single reservoir entirely into work which is a violation of the Kelvin-Planck statement

No questions assigned to the following page.

4) 6.17.14

$n_{rev} = 1 - \frac{T_C}{T_H}$, so $n_{rev} = 1 - \frac{400}{1200} = 0.667 = 66.7\%$

a) Actual efficiency: $n = \frac{W}{Q_H} = \frac{450}{900} = 0.5 = 50\%$

$n_{act} < n_{rev}$, so process is irreversible

b) Work Done: $W = Q_H - Q_C = 900 - 300 = 600 kJ$

Actual Efficiency: $n = \frac{W}{Q_H} = \frac{600}{900} = 0.667 = 66.7\%$

$n_{act} = n_{rev}$, so process is reversible

c) Finding Q_H: $W + Q_C = 600 + 400 = 1000 kJ$

Actual Efficiency: $n = \frac{W}{Q_H} = \frac{600}{1000} = 0.60 = 60\%$

$n_{act} < n_{rev}$, so process irreversible

d) $n_{act} = 75\%$ isn't possible since $n_{act} > n_{rev}$ isn't allowed

5a) 6.17.17

Thermal Efficiency: $n = 1 - \frac{T_C}{T_H} = 1 - \frac{400}{1600} = 0.75 = 75\%$ efficiency

5b) First, convert temps to Kelvin:

$T_H = 500°C = 773K$ and $T_C = 20°C = 293K$

Finding Efficiency: $n = 1 - \frac{T_C}{T_H} = 1 - \frac{293}{773} = 0.621 = 62.1\%$

Thermal Efficiency: $n = \frac{W_{cycle}}{Q_H}$, so find Q_H

$Q_H = \frac{W_{cycle}}{n}$, so $Q_H = \frac{1000}{0.621} = 1610.31 kJ$

Finding Q_C: $Q_C = Q_H - W$, so $Q_C = 1610.31 - 1000 = 610.31 kJ$

5c) $T_C = 40°F = 499.67R$

Using efficiency: $n = 1 - \frac{T_C}{T_H}$, so $0.6 = 1 - \frac{499.67}{T_H}$

So $0.4 = \frac{499.67}{T_H}$, so $T_H = \frac{499.67}{0.4} = 1249.18R$

$T_H = 1249.18R = 789.51°F$

No questions assigned to the following page.

5d) $T_H = 727°C = 1000K$

Efficiency: $n = 1 - \frac{T_C}{T_H}$, so $0.40 = 1 - \frac{T_C}{1000}$, so $T_C = 600K$

$T_C = 600K = 327°C$

6) 6.17.18

$T_H = 1100°C = 1373K$

$T_C = 15°C = 288K$

Efficiency: $n = 1 - \frac{288}{1373} = 0.7902 = 79\%$

The calculated efficiency is the maximum theoretical efficiency

No real power cycle can reach the max theo efficiency, so his claim is likely not correct, since it's practically impossible.